

Literature-Implied Answer to a Dotted Subideal Inclusion in arXiv:1507.03241

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Status

This note records a later-literature answer to the dotted-arrow question in Section 5 of Wallis, arXiv:1507.03241. The answer comes from Freeman, Schlumprecht, and Zsak, arXiv:1612.01153.

The packet is a literature-status record, not a new theorem. The relationship is agent-identified from the supporting theorem and the remark following its proof, so the status is *literature-implied answer*.

Source Signal

Wallis summarizes the then-known closed subideal structure of $\mathcal{L}(\ell_p, c_0)$, $1 < p < \infty$, by the diagram

$$\{0\} \Rightarrow \mathcal{K} \Rightarrow [\mathcal{G}_{I_{p,0}}] \cdots \triangleright \mathcal{FSS} \dashrightarrow \mathcal{L}.$$

The source explicitly says that a dotted arrow denotes an inclusion which is not known to be proper.

The same section then proves a stronger diagram only for the special range $1 < p < 2$, inserting further ideals between $[\mathcal{G}_{I_{p,0}}]$ and $\mathcal{FSS}$. The dual statement gives the analogous information for $\mathcal{L}(\ell_1, \ell_q)$ when $2 < q < \infty$.

Supporting Theorem

Freeman, Schlumprecht, and Zsak prove in arXiv:1612.01153 that for every $1 < p < \infty$ the space $\mathcal{L}(\ell_p, c_0)$ has continuum many closed ideals. In Theorem 1, they construct continuum many closed ideals between $\mathcal{J}^{I_{U,V}}$ and $\mathcal{FSS}$.

The remark following the proof of Theorem 1 identifies the base ideal in the range $2 \leq p < \infty$: in that range, $\mathcal{J}^{I_{\ell_p, c_0}}$ and $\mathcal{J}^{I_{U,V}}$ are identical. Hence their continuum family lies above $[\mathcal{G}_{I_{p,0}}]$ and below $\mathcal{FSS}$, proving that the dotted inclusion from Wallis's diagram is proper for $2 \leq p < \infty$.

Combining this later result with Wallis's own $1 < p < 2$ result answers the dotted-inclusion question for all $1 < p < \infty$.

Scope

This note does not address the open problems left at the end of arXiv:1612.01153, including whether the lattice embedding preserves meets or whether $\mathcal{FSS} \subset \mathcal{J}^K$ in their Theorem 2.

References

1. B. Wallis, *Closed ideals in $\mathcal{L}(X)$ and $\mathcal{L}(X^*)$ when X contains certain copies of ℓ_p and c_0* , arXiv:1507.03241.
2. D. Freeman, Th. Schlumprecht, and A. Zsak, *Closed ideals of operators between the classical sequence spaces*, arXiv:1612.01153.